

# The Price of Realism: A Filtration of the State Space and Three Studies of Its Transitions

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## Abstract

Finitely additive coherence is what observation natively supplies;  $\sigma$ -additivity and value-definiteness are further *representation properties* — signatures of a world, not of the data. This umbrella note reads three companion papers as one classification of the price of those signatures. The organising object is a filtration of the state space,  $\mathcal{S} \supseteq \mathcal{S}_\sigma \supseteq \mathcal{S}_{\text{df}}$ , whose three transitions — *pasting*, *regularity*, *sharpness* — are introduced in the companion paper on distributivity and realism. That paper studies the sharpness transition (distributivity versus value-definiteness, Kochen–Specker); the paper on a  $\sigma$ -essential contextual state studies the regularity transition (finite versus countable additivity, resolved by a witness constructed in ZFC at  $\omega_1$ ); and the paper on commensurability of observation protocols studies the pasting transition (local coherence versus global realisability, with a taming catalogue and an odd-ring separating witness). We tabulate the three transitions against the coordinate in which each locates its boundary — algebraic, set-theoretic, dynamical — and record what is genuinely common: the observer-first reading, the recurring appearance of non-distributivity as the tame/wild discriminator, and a method (pre-registered predictions, witness-grade verdicts, machine-checked lemmas). We are careful about what is *not* yet established across the corpus: whether the papers’ local coherence notions coincide as one notion at different rungs, and whether a compactness “quarantine” provably separates the two constructed witnesses into disjoint regimes, are stated here as the corpus’s open bridging questions, not as settled facts. This note references the three papers; it does not absorb them.

## 1 The observer-first reading

An observer running a protocol obtains, natively, a family of finite relative frequencies that cohere across overlapping views. That is all observation supplies. Two further properties are routinely assumed of the state such data “comes from” — that it is countably additive, and that it assigns definite truth values — but neither is given by the data. They are *representation properties*: signatures of the world being observed and of the protocol used to observe it, not of empirical adequacy itself. The programme these three papers form is the price list for those signatures: what structure a world’s language and an observer’s protocol must have before countable additivity and value-definiteness come for free, and — in two cases settled by explicit construction — proofs that outside that structure the signatures can be genuinely absent while coherence remains.

This note is an umbrella. It states the shared object, tabulates how the three papers divide the labour, records the corpus-level commonalities and the genuinely corpus-level open questions, and points to each paper for the mathematics. It deliberately does not merge them: each paper travels to its own audience, and the spine carries only what no single paper can say alone.

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## 2 The filtration and its three transitions

The companion paper on distributivity and realism introduces the organising object: on the state space of an observation algebra there is a natural filtration

$$\mathcal{S} \supseteq \mathcal{S}_\sigma \supseteq \mathcal{S}_{\text{df}},$$

from all coherent states, to the  $\sigma$ -additive ones, to the dispersion-free (value-definite) ones. Its three inclusions are transitions of different mathematical character, named there:

- (1) *pasting* — from locally coherent window-data up to a single globally realising state ( $\text{EA} \rightarrow \text{PR}$ );
- (2) *regularity* — from a finitely additive global state up to a countably additive one ( $\text{PR}_{\text{f.a.}} \rightarrow \text{PR}_\sigma$ );
- (3) *sharpness* — from a  $\sigma$ -additive state up to a value-definite one ( $\text{PR} \rightarrow \text{VDR}$ ).

Distributivity collapses the filtration — on a Boolean algebra all three transitions are free — and non-distributivity can open any of them. The corpus-level observation this note records is that **each companion paper is the deep study of exactly one transition**, and that the three locate the boundary of their transition in three different coordinate systems.

| Transition | Paper                    | Boundary (tame side)           | Separating object (wild side)              |
|------------|--------------------------|--------------------------------|--------------------------------------------|
| pasting    | commensurability         | acyclic reporting; the tamings | odd-ring half-model (a golden-mean PR-box) |
| regularity | $\sigma$ -essential      | Boolean; Polish-representable  | the product Ulam carrier, in ZFC           |
| sharpness  | distributivity & realism | distributivity                 | Kochen–Specker (imported, reframed)        |

Table 1: The three transitions of the filtration, the paper that studies each, the structural boundary separating its tame from its wild side, and the object witnessing the wild side. Two of the three witnesses — the odd-ring half-model and the product Ulam carrier — are constructed in the companion papers rather than imported.

The table is the weave. One question — *when does empirical adequacy purchase realism?* — is answered as a classification, with the boundary located algebraically (distributivity), set-theoretically (beyond Polish, at  $\omega_1$ ), and dynamically (odd recurrence). Two of the separating witnesses are new constructions; the third (Kochen–Specker) is classical, reframed as the sharpness-transition obstruction.

## 3 What is genuinely common

Three things hold across all three papers and belong to none of them alone.

**Non-distributivity as the discriminator.** In each transition the tame side is, at root, a distributive (or distributively-representable) regime, and the wild side is entered through non-distributivity: Kochen–Specker needs  $\dim H \geq 3$ ; the  $\sigma$ -essential witness is provably not a lattice; the pasting failures are carried by non-distributive constraint geometry. That non-distributivity is the tame/wild discriminator in *every* transition is a corpus-level pattern no single paper isolates.

**The observer-first price list.** Section 1: the three papers are, together, the statement that  $\sigma$ -additivity and value-definiteness are representation properties with a computable price, and two proofs that the price is real (the two constructed witnesses).

**Method.** The corpus shares a working method, stated once here without ceremony: predictions are pre-registered before testing; a verdict of “realisable/tame” is admitted only from a theorem or an exact enumeration, of “contextual/wild” only from an exhibited witness; and load-bearing combinatorial lemmas are machine-checked (the  $\sigma$ -essential main theorem and the commensurability paper’s layer-injectivity discriminant are certified in Lean 4).

**The novelty ledger, honestly.** The classical mathematics each paper uses is credited there as classical (Gleason and Kochen–Specker for sharpness; Derr–Williamson, Ulam, and the Boolean baseline for regularity; Wainwright–Jordan, Vorob’ev, De Loera–Onn and stable-set theory for pasting). The corpus’s own contribution is the observer-first framing, the filtration as its organising object, and two constructed witnesses where the literature had only imported or absent ones. This is a bridge over known mathematics with two new construction points; the umbrella claims no more.

## 4 The corpus-level open questions

Two bridging claims would strengthen the weave from a tabulation into a single theorem. Neither is established; we state them as the corpus’s open problems, and flag that asserting either prematurely would misrepresent the corpus.

**Conjecture 4.1** (one coherence notion across rungs). *The local coherence notions of the three papers — the commensurability paper’s EA (pairwise agreement on common subalgebras), the  $\sigma$ -essential paper’s finite coherence (extendability to a global finitely additive two-valued state), and the top rung  $\mathcal{S}$  of the filtration — are one notion evaluated at different rungs.*

**Remark 4.2** (status of Conjecture 4.1). *As stated in the three papers* these notions are not literally identical: EA is a pairwise-local condition, finite coherence is a global finite-additive extendability condition, and they are known to coincide only under further hypotheses. The papers exhibit the same *phenomenon* — coherence that fails to globalise — at adjacent rungs, which is a genuine rhyme; whether it is a single notion (transfer) rather than three instances of a family (rhyme) is open, and the corpus should claim the rhyme, not the identity, until the coincidence is proved.

**Conjecture 4.3** (the quarantine hinge). *The two constructed witnesses inhabit provably disjoint regimes: a compactness argument (finite-alphabet delay logics are compact/clopen, so clopen charges are measures and finite certificates suffice by Farkas) shows the  $\sigma$ -essential pathology cannot occur in the dynamical (finite-alphabet) regime where the pasting witness lives. If so, the corpus does not merely exhibit two failures — it proves they are sorted, upgrading the tabulation to a classification.*

**Remark 4.4** (status of Conjecture 4.3). The compactness/Farkas argument is currently a sketch (a roadmap “Layer-1 licensing” result in the commensurability programme), not an audited proof. Until it is settled, the spine’s central architectural claim — that the two witnesses live in provably disjoint worlds, so the failures are classified rather than merely collected — is a conjectural hinge. We state it as the corpus’s most valuable next lemma precisely because it is what would make the weave a theorem.

A third open thread is shared and worth stating uniformly: in each transition, *latticehood* appears to force liftability (and hence tameness) — the orthomodular-lattice form of each paper’s boundary question is conjectured to collapse to the tame side. Whether these are three phrasings of one conjecture ( $\text{latticehood} \Rightarrow \text{liftability} \Rightarrow \text{tameness}$ ) is itself part of Conjecture 4.1.

## 5 Reading and dissemination order

For a reader: distributivity & realism first (it introduces the filtration), then  $\sigma$ -essential (the deepest single result), then commensurability (the broadest). For dissemination the crown result travels first and alone: the  $\sigma$ -essential paper is self-contained and Lean-certified, and its Derr–Williamson cell-closure framing does not depend on the other two. The commensurability paper follows; the distributivity paper may belong to a different venue class (foundations of physics) than the two mathematics papers. The spine lets each paper travel light while carrying, itself, only the corpus-level content: the filtration as shared object, the observer-first moral, and the two open bridging questions of §4.